

Adaptation Utility and Transactional Safety in Open-World Recognition Systems

A Study of Why Qualified Recognition, Discovery, and Adaptation
Stages Fail to Compose

by

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Approval

This thesis-style report has been prepared for academic review. Institutional approval language, degree title, department name, supervisor names, signatures, and submission date must be replaced with the requirements of the submitting university before formal deposit.

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Declaration

I declare that this report presents the research and experimental record of the GEODE project as described in the cited repository artifacts. All claims in the report are bounded by the preregistered protocols, completed experiments, and immutable evidence indexes. Any institutional declaration required by the degree-awarding university should replace this provisional text before submission.

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Abstract

Open-world recognition systems must do more than reject unfamiliar inputs. They must accumulate rejected evidence, identify persistent candidate groups, request semantic supervision, integrate confirmed classes, route future inputs, and preserve auditability and rollback. Research commonly evaluates these stages independently, but strong stage-level performance does not imply that their outputs contain the information required by downstream stages.

This thesis investigates stage composition in GEODE, an experimental framework for explicit geometric class models and transactional model lifecycles. The realized experiments use frozen 384-dimensional DINOv2 features on bounded CIFAR-10 open-world episodes. The research progresses through three questions. First, can explicit geometric heads compete with standard discriminative controls under a frozen representation? Second, can independently qualified rejection, discovery, confirmation, adaptation, and routing mechanisms compose into a useful lifecycle? Third, does replacing stage metrics with post-integration adaptation utility resolve the composition failure?

The predictive study found that a weighted rank-32 affine head reached 91.73% balanced accuracy, remaining 5.03 percentage points behind a 96.77% RBF SVM control. The open-world stage study retained a rank-32 class-conditional Gaussian with 92.08% known coverage, 86.17% unknown recall, 0.9561 AUROC, and 98.67% review precision. HDBSCAN reached 100% stage-wise group recall and review precision, while confirmed rank-16 insertion improved isolated new-class success by 43.33 points. These stages nevertheless failed to compose: grouped loops integrated 0 of 3 confirmable classes.

The successor protocol registered adaptation utility—the change in balanced accuracy on known and discovered classes after integration—as the primary endpoint at 50 reviewed labels per episode. Density-core review sets covered only 34.0–39.1% of the rejected region under a nearest-neighbor support diagnostic, compared with 99.37–99.78% for boundary-inclusive sets. Across nine seed-by-arrival cells, utility selection beat core selection in all nine by 3.593 points on average, with a paired 95% bootstrap interval of [2.794, 4.382]. The result was below the preregistered 5-point practical threshold, and six cells violated the 2-point remaining-unknown-recall

band. Parent-scoped residual fusion repaired a prior locality defect, preserving 100% of unaffected predictions, but added at most 0.204 points of utility.

The final outcome is a rigorous negative systems result. Pure discovery cores are insufficient statistics for adaptation; broader support improves utility; but representativeness alone does not preserve the remaining unknown space. The thesis contributes an adaptation-utility endpoint, typed stage-interface contracts, deterministic episode replay, confirmation-gated transactions, exact rollback, and artifact-only verification. It concludes that open-world recognition must be evaluated as a jointly constrained lifecycle rather than as a sequence of independently optimized benchmarks.

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List of Abbreviations

AUROC	Area under the receiver operating characteristic curve
BA	Balanced accuracy
CSG	Constructive solid geometry
DDU	Deep Deterministic Uncertainty
EVM	Extreme Value Machine
GCD	Generalized category discovery
GEODE	Geometric, explicit, open-world development environment used in this work
NLL	Negative log likelihood
OOD	Out-of-distribution
SDF	Signed distance field

Chapter 1

Introduction

1.1 Motivation

Most classification benchmarks assume a closed world: training and evaluation share a fixed class inventory, and every input belongs to one of those classes. Deployed systems rarely enjoy that assumption. New products, species, documents, faults, behaviors, and domains appear after training. A system that must always choose a known label can become confidently wrong precisely when human attention is most valuable.

Open-set recognition addresses part of this problem by allowing rejection. Open-world recognition extends the objective: a system should identify known classes, reject unfamiliar inputs, acquire supervision for novel classes, and incrementally expand its knowledge [1]. This formulation is operationally demanding. Rejection creates an unresolved observation, not a new class. The system must decide what to retain, which rejected examples form a coherent candidate, which examples deserve limited review, how confirmation authorizes mutation, how a child model changes acceptance thresholds, and how the change can be reversed.

Each stage has familiar local metrics. Rejection is summarized by AUROC, known coverage, and unknown recall. Discovery is summarized by cluster purity, adjusted Rand index, or category recall. Adaptation is summarized by target accuracy or forgetting. Routing is summarized by top- k inclusion and compute reduction. These metrics are useful diagnostics, but they do not answer the system-level question: did the reviewed evidence create a better model without damaging what the system still does not know?

1.2 Research problem

The central problem of this thesis is *interface sufficiency*. A producer stage may optimize a metric while discarding statistics required by its consumer. A clusterer can produce a pure density core that is easy to label, while an adapter needs support over the class boundary and low-density modes. A router can include the correct expert almost always, while hiding exactly the unknown cases that require exhaustive fallback. A transaction can roll back exactly while its forward edit changes predictions outside the declared region.

The thesis therefore treats open-world recognition as a constrained lifecycle whose interfaces are explicit experimental objects.

1.3 Research questions

The work addresses four research questions:

RQ1: Can compact explicit geometric class artifacts approach the predictive quality of matched discriminative controls under a frozen deep representation?

RQ2: Do independently qualified rejection, discovery, human confirmation, adaptation, and routing stages compose into a review-efficient open-world loop?

RQ3: Does selecting reviewed evidence for downstream adaptation utility, rather than density-core purity, improve post-integration performance at an equal human-review budget?

RQ4: Can transactional controls—typed interfaces, confirmation lineage, edit locality, exact replay, and rollback—make lifecycle failures auditable without overstating predictive success?

1.4 Thesis statement

The thesis advances the following bounded statement:

Stage-wise qualification is insufficient for open-world lifecycle qualification. At a fixed review budget, representative review selection improves adaptation utility over density-core selection, but safe lifecycle advancement additionally requires explicit preservation of known performance, remaining-unknown recall, lineage, locality, and rollback.

This statement does not claim closed-set parity, autonomous semantic class creation, SDF necessity, authoritative learned routing, or successful end-to-end qualification.

1.5 Contributions

The research makes five contributions:

1. A controlled comparison separating the predictive value of explicit geometry from its auditability and editability.
2. A preregistered stage-to-system study showing that strong rejection, discovery, and isolated adaptation results can fail to compose.
3. A registered adaptation-utility endpoint at a fixed review budget, with known-class and remaining-unknown safety constraints.
4. A quantified sufficient-statistics diagnosis showing that density cores are pure but geometrically unrepresentative in the realized episodes.
5. A transactional evaluation harness with typed interfaces, immutable lineage, deterministic replay, human-confirmation gates, local edit scopes, and exact rollback.

1.6 Organization

Chapter 2 reviews open-world recognition, discovery, routing, explicit geometry, and lifecycle safety. Chapter 3 formalizes the problem and endpoint. Chapter 4 presents the GEODE architecture. Chapter 5 records how the methodology evolved, including failed hypotheses and the tests performed. Chapters 6 and 7 describe the v7 and v8 experimental protocols. Chapter 8 reports results. Chapter 9 discusses the findings and answers the research questions. The remaining chapters cover validity, ethics, reproducibility, and conclusions. Appendices record gates, branch dispositions, and artifact commands.

Chapter 2

Background and Related Work

2.1 Open-set and open-world recognition

Bendale and Boulton formalized open-world recognition through explicit rejection and incremental class addition [1]. Their Nearest Non-Outlier method controls open-space risk using metric distance to class means. OpenMax later calibrated deep activation vectors using extreme-value theory [2]. The Extreme Value Machine models Weibull tails over class margins, yielding incrementally updatable class boundaries [3].

Feature-space baselines remain strong when representations are frozen. Maximum softmax probability is a common baseline [16]. Mahalanobis scoring uses class-conditional feature distributions [17]; DDU combines deterministic features with density estimation [18]; and deep nearest-neighbor distance can provide strong OOD detection without a parametric head [19]. DUQ and SNGP introduce distance awareness into discriminative models [20, 21].

These methods play a role similar to the SDF in the original GEODE motivation: they define a calibrated acceptance region with an outside signal. The literature suggests that the mechanism should be treated as substitutable. This thesis follows that implication by retaining a low-rank Gaussian when it outperforms geometric support.

2.2 Streaming novelty and emerging classes

The data-stream literature operationalized novelty buffering and class evolution before modern self-supervised representations. ECSMiner handles classification and novel-class detection under drift [4]. SAND and ECHO address scarce labels and evolving concepts [5, 6]. MINAS maintains micro-cluster summaries and recurring novelty [7], while SENCForest targets streaming emerging classes [8].

This lineage demonstrates that reject-buffer-group-update loops are not novel

in concept. The open problem examined here is the composition of modern frozen features, explicit human publication authority, transactional adaptation, and cross-model support routing under one measured endpoint.

2.3 Novel and generalized category discovery

Generalized category discovery clusters unlabeled data containing known and novel classes, often over strong self-supervised features [9]. ORCA jointly learns seen and novel categories in open-world semi-supervised learning [10]. Grow and Merge and incremental GCD extend discovery across episodes [11, 12].

Clustering mechanisms include DBSCAN [22], HDBSCAN [23], and FINCH [24]. Purity and category recovery are natural discovery metrics, but they do not measure whether a small reviewed subset supports a downstream class model. The present study isolates this difference.

2.4 Continual learning and modular routing

Continual learning studies acquisition under stability-plasticity constraints. The present work uses a narrower transactional setting: the representation is frozen, semantic mutation requires confirmation, and every child has an exact parent.

Expert Gate routes examples among task experts using reconstruction error [13]. DEMix and Branch-Train-Merge provide modern modular precedents [14, 15]. In GEODE, fingerprints establish structural compatibility, while empirical profiles estimate support. A profile may only become authoritative if it agrees with exhaustive inference and fails safely on unknown inputs.

2.5 Explicit geometry and low-rank class models

Signed distance fields and constructive solid geometry provide editable spatial representations [25, 26]. Their attraction in feature space is governance: an operator could inspect components, remove a lobe, merge regions, or roll back an edit. Predictive competitiveness, however, is a separate empirical question.

The retained affine and Gaussian components use a center, orthonormal tangent basis, tangent variances, and residual variance. This construction relates to probabilistic PCA and mixtures of factor analyzers [27, 28]. Proper likelihood includes both normalized distance and volume terms; raw geometric support does not automatically yield cross-class comparability.

2.6 Research gap

Prior work validates each stage individually. What is uncommon is an end-to-end study in which:

- review cost is fixed;
- stage interfaces and sufficient statistics are registered;
- semantic mutation requires linked confirmation;
- child thresholds and routes carry lineage;
- exact rollback and locality are independently tested; and
- the primary endpoint is post-integration utility.

This thesis addresses that systems-level gap. Its novelty claim concerns the controlled composition study and audit boundary, not the existence of the discover-label-update loop or any individual mechanism.

Chapter 3

Problem Formulation

3.1 Open-world episode

Let $f(x) \in \mathbb{R}^d$ be a frozen representation. At the beginning of episode e , parent model M_e recognizes class set $\mathcal{K}_e$. The stream contains known inputs and an arriving class $c_e \notin \mathcal{K}_e$; a separate set $\mathcal{R}_e$ remains unknown after the episode.

The rejector maps an input to a known class or rejection. Rejected records enter a bounded buffer. A discovery policy forms a persistent review candidate. A review selector chooses at most B examples. An authorized confirmation assigns semantic meaning, after which an adapter may publish child M_{e+1} . A router chooses compatible bundles or falls back to exhaustive evaluation.

3.2 Semantic states

Three states must remain distinct:

Rejected: an acceptance score crosses a threshold.

Grouped: rejected examples exhibit persistent structure.

Confirmed: an authorized review assigns semantic meaning.

Only confirmation authorizes a semantic class or model mutation. Group IDs are persistent review objects, not provisional semantic labels.

3.3 Adaptation utility

The v8 primary endpoint is episode utility:

$$U_e = \text{BA}_{\mathcal{K}_e \cup \{c_e\}}(M_{e+1}) - \text{BA}_{\mathcal{K}_e \cup \{c_e\}}(M_e). \quad (3.1)$$

The parent receives zero credit on the unavailable arriving class. This avoids the misleading comparison in which a no-update parent is evaluated only on classes it already knows.

The cumulative endpoint after N episodes is

$$U_{1:N} = \text{BA}_{\mathcal{K}_1 \cup \{c_1, \dots, c_N\}}(M_{N+1}) - \text{BA}_{\mathcal{K}_1 \cup \{c_1, \dots, c_N\}}(M_1). \quad (3.2)$$

3.4 Constraints

Utility is evaluated subject to:

$$\text{reviews}_e \leq B, \quad (3.3)$$

$$\text{BA}_{\mathcal{K}_e}(M_e) - \text{BA}_{\mathcal{K}_e}(M_{e+1}) \leq \epsilon, \quad (3.4)$$

$$\text{Recall}_{\mathcal{R}_e}(M_e) - \text{Recall}_{\mathcal{R}_e}(M_{e+1}) \leq \delta. \quad (3.5)$$

The registered values are $B = 50$, $\epsilon = 0.01$, and $\delta = 0.02$. Transactions must also satisfy confirmation, graph, lineage, replay, rollback, and routing-fallback invariants.

3.5 Interface sufficiency

For adjacent stages $P \rightarrow C$, let S_C be the sufficient statistics needed by consumer C and E_P be the evidence emitted by producer P . An interface mismatch exists when

$$S_C \not\subseteq E_P. \quad (3.6)$$

The central candidate mismatch is clusterer-to-review: the producer emits dense core members and persistence, while the adapter needs support over modes, covariance directions, and parent-boundary regions. The experiments compare review policies that alter E_P at fixed cost.

3.6 Registered decisions

The pivotal utility-selected policy advances only if its mean gain over core selection is at least 5 points, the paired 95% bootstrap lower bound is above zero, at least 7/9 cells are positive, and all safety and transaction constraints pass. The 5-point operand is a practical effect threshold. It is not replaced by a positive confidence interval after results are known.

Chapter 4

The GEODE System

4.1 Design objectives

GEODE separates five concerns:

1. frozen representation;
2. class-conditional support geometry;
3. calibrated decision or rejection;
4. versioned lifecycle operations; and
5. evidence and reproducibility.

This separation allows a head to fail predictive parity while its transaction machinery remains useful. It also permits a Gaussian or kNN mechanism to replace an SDF without weakening a systems claim.

4.2 Explicit class components

For a component with center μ , orthonormal tangent basis $V \in \mathbb{R}^{d \times r}$, tangent variance vector λ , and isotropic residual variance σ^2 , define

$$q(x) = \sum_{j=1}^r \frac{(v_j^\top (x - \mu))^2}{\lambda_j} + \frac{\|(I - VV^\top)(x - \mu)\|_2^2}{\sigma^2}. \quad (4.1)$$

The corresponding Gaussian log likelihood is

$$\log p(x | k) = -\frac{1}{2} \left[q(x) + \sum_{j=1}^r \log \lambda_j + (d - r) \log \sigma^2 + d \log(2\pi) \right]. \quad (4.2)$$

The volume terms are essential for cross-class comparison. A raw field that captures coverage can still be poorly calibrated for discrimination.

4.3 Immutable model bundles

A bundle records the representation identity, class order, components, readout, thresholds, router profile, parent hash, and schema version. Publication creates an immutable child rather than mutating the parent in place. Rollback changes the current pointer to the exact parent while retaining the full append-only history.

4.4 Review and confirmation

Rejected observations carry stable record IDs and representation lineage. Discovery produces persistent review IDs. Confirmation records reference those reviews and contain the authorized outcome. The adapter rejects missing, stale, or mismatched confirmation. This design prevents a coherent unlabeled cluster from silently becoming a semantic class.

4.5 Typed interfaces

The v8 protocol registers five interfaces:

1. rejector to buffer;
2. buffer to clusterer;
3. clusterer to review selector;
4. review to adapter; and
5. adapter to router.

Seven versioned schemas encode endpoint evidence, episode replay, interface audits, threshold transfer, review selection, residual scope, and integrated routing. Six negative cases verify fail-closed behavior for leakage, stale class order, threshold mismatch, budget overflow, missing confirmation, and rollback-parent drift.

4.6 Research progression

The complete program is cumulative. Early studies tested primitive fitting, constructive operations, calibrated readouts, precision policies, scaling, and transactions. The thesis focuses on the progression most relevant to its questions:

v6/v6.1: predictive competitiveness of explicit heads and lifecycle locality;

v7: stage-wise open-world qualification and integrated composition;

v8: adaptation utility, interface statistics, equal-budget review selection, and parent-scoped residuals.

Every milestone uses immutable parents and registered advancement gates. Failed gates close downstream branches rather than being silently redefined.

Chapter 5

Evolution of the Research Methodology

5.1 Why the methodological history matters

The final adaptation-utility protocol was not the initial methodology. It emerged after a sequence of increasingly restrictive experiments exposed different failure modes. Recording only the final protocol would hide why its endpoint, controls, and stopping rules exist. This chapter therefore separates the *research process* from the final experimental methods.

The progression can be summarized as:

$$\begin{aligned} \text{implementation correctness} &\rightarrow \text{geometric expressiveness} \\ &\rightarrow \text{predictive competitiveness} \\ &\rightarrow \text{stage qualification} \\ &\rightarrow \text{lifecycle utility.} \end{aligned} \tag{5.1}$$

Each transition occurred because the previous level proved necessary but insufficient. The methodology did not advance by treating every experiment as a success. Null, negative, blocked, and infeasible outcomes determined which questions were legitimate to ask next.

5.2 Phase I: implementation correctness and semantic tests

The recorded methodology begins with the repaired Tier-6 temporal baseline and M0 protocol lock, followed by M1 semantic correctness. The earliest methodology focused on whether the implemented geometric operations meant what their formulas and

APIs claimed. These tests preceded performance comparisons because a predictive result is uninterpretable when the primitive semantics are wrong.

The test families included:

- seeded-RNG ownership tests, which removed dependence on ambient NumPy state;
- normalized-softmin conservation tests, which detected an exact $\log(4)/2$ field shift under pruning;
- finite-difference gradient tests for hard constructive operations;
- scalar-versus-vectorized optimizer parity;
- CPU-versus-OpenCL numerical parity;
- serialization and constructor smoke tests; and
- numerical audits of approximate ellipsoid distance.

The ellipsoid audit established an important claim boundary. The implemented radial correction is exact for spheres but can deviate substantially for eccentric ellipsoids. Terminology was therefore narrowed from an exact metric distance claim to an implemented first-order correction. This pattern—test a semantic claim, then restrict the claim when the test fails—became a core methodological rule.

5.3 Phase II: M0–M4 baseline construction and causal ablation

M0 froze experiment configurations, dataset manifests, array fingerprints, seeds, split policies, resource accounting, and deterministic smoke artifacts. The purpose was to make later comparisons traceable to identical inputs.

M2 then decomposed geometric fitting from decision readout. Geometry and calibration used disjoint data. Five readouts were compared on the same fitted artifact. Temperature scaling left accuracy unchanged at 84.69% while reducing NLL from 1.48 to 0.50 and ECE from 0.60 to 0.03; multinomial readout added 0.19 accuracy points and reduced NLL to 0.49. This established that field quality and probability calibration had to be measured separately.

M3 introduced eight matched classical baselines over identical transformed features and split hashes. Over five seeds, logistic regression reached 84.02%, RBF SVM 84.00%, and the GEODE multinomial readout 83.11%. The methodology thereafter

required same-representation controls rather than comparing against unrelated published numbers.

Early causal ablations examined additive geometry, held-out subtractive excision, and active repair. Subtractive constructive geometry showed null or slightly negative aggregate effects, and active repair generated no accepted candidates. The methodology therefore stopped treating more expressive CSG as automatically beneficial. Subtraction remained opt-in rather than becoming a default source of unmeasured complexity.

The testing strategy expanded from unit semantics to:

- matched ablations with identical representations and budgets;
- seed-level and pooled paired intervals;
- corruption and leakage fixtures;
- runtime, memory, and serialized-state accounting; and
- append-only evidence manifests.

5.4 Phase III: M5–M7 stress, sequence, and uncertainty studies

M5 asked whether alternative high-dimensional fitters improved robustness under fixed budgets. Seven synthetic fitters passed through a synthetic gate, a five-seed Tier-4 screen, and a locked Tier-5 confirmation. A shrinkage candidate gained 0.75 points on the selection screen but lost 1.56 points on confirmation, used 4.4 times as many primitives, and took 3.2 times longer. The current fitter was retained. This result motivated confirmation stages and resource co-primary metrics.

M6 tested whether explicit geometric state could represent temporal context. Exact, reservoir, ensemble, and hybrid causal representations were compared through periodic synthetic gates, train-only tuning, and a locked 50,000/10,000 WikiText episode. GEODE reached 30.36%, below a linear model at 34.64%, a matched 5-gram at 44.50%, and a practical 5-gram at 47.61%. The result established representational flexibility, not competitive temporal learning.

M7 expanded evaluation beyond top-1 accuracy to controlled shift, selective prediction, calibration, conformal diagnostics, and real CIFAR-100/SVHN OOD families. Maximum-probability AUROC reached 0.786 and 0.831 and exceeded the tested Mahalanobis and kNN scores, but FPR95 remained 0.579 and 0.431. The claim was restricted to relative ranking under the frozen fit.

5.5 Phase IV: M8–M10 editability, scaling, and open-set inference

M8 combined corruption tests, seed-stability tests, explicit insertion and deletion, rollback, and online updates. Broad robustness was rejected, while predictions remained 93.0% seed-agreeing despite primitive drift. Behavioral edit gates showed that insertion, deletion, and rollback could be local and reversible in the tested setting.

M9 measured editability rather than assuming that compact artifacts scaled. One-factor sweeps varied class count, dimension, and primitives per class over five repeats with operation counters. Locality gates passed and insertion stayed below 0.18 ms, but exhaustive inference and dense 128-dimensional snapshots exposed bounded scaling costs.

M10 introduced support profiles, abstention, and leave-class-out selection in a locked three-episode toy matrix. No score, threshold, and geometry combination jointly reached 90% known coverage and 50% unknown recall across all episodes. The experiment showed that calibrated closed-set scores did not automatically produce reliable open-set inference.

5.6 Phase V: M11 discovery methodology

M11 was not one experiment but a long methodological sequence on accumulated unsupported observations. It began with bounded evidence, automatic clustering, stable review IDs, and confirmation-gated adaptation. Initial studies exposed category merging and fragmentation, after which the sequence tested:

- DBSCAN flag budgets and radius selection;
- HDBSCAN under variable-density hypotheses;
- PCA versus normalized raw MobileNet representations;
- corrected evaluation on the final accumulated partition rather than the first review snapshot;
- larger accumulation windows;
- FINCH without a supplied group count;
- DenStream-style fading micro-clusters;
- silhouette-selected k -means;
- transductive joint PCA;

- delayed must-link and cannot-link metric learning;
- COBRAS-style split and merge feedback;
- answer-blind active pair queries;
- missing, flipped, and contradictory-answer stress tests;
- independent confirmation of pairwise answers;
- 450-observation lower-tail reliability studies at each error rate; and
- a ResNet-18 representation control.

The evaluation boundary itself evolved. Early group metrics used first-review snapshots, which did not match the operational goal. M11.26 separated online review delay from autonomous partition quality and evaluated the complete flagged buffer, including noise. Under the corrected protocol, the retained normalized-MobileNet HDBSCAN policy reached 61.1% distinct-group recall, 79.8% purity, and full two-group recovery in 4/9 cells, but noise-inclusive adjusted Rand index was only 0.059.

Feedback studies showed positive mean improvements but weak lower-tail safety. At 12.5% respondent error, independently confirmed constraints still accepted harmful constraints in 10.44% of observations; fifth-percentile ARI change was negative. Constraints therefore remained analysis-only and unpersisted. M11.40 closed the sequence by retaining only the automatic review-grouping policy and prohibiting autonomous mutation.

5.7 Phase VI: M12 routing and scaling methodology

M12 began with one-factor synthetic sweeps over classes, dimensions, primitive counts, and batch size. Exhaustive inference exhibited approximately linear latency growth in class and primitive count. Subsequent iterations tested:

- conservative support-sphere lower bounds with exact early stopping;
- grouped active sample-class evaluation;
- incumbent initialization;
- nearest-neighbor indexes over support centroids;
- greedy primitive removal;

- disjoint confirmation streams for tentative pruning; and
- a complete exit audit with counters, agreement, fallback, and latency.

Some candidates preserved exact predictions, but they either evaluated too much of the exhaustive class set, failed normalized-score requirements, or did not improve tail latency. The methodology therefore rejected operation-count improvement as sufficient evidence for production routing.

5.8 Phase VII: M13–M15 primitive and field semantics

M13 reopened primitive-family comparisons under matched full-covariance, diagonal-covariance, and trace-matched spherical fits, including primitive-aware OpenCL paths and the full additive/excision/repair matrix. Spherical additive geometry reached 83.80%, 0.463 points above diagonal, but subtraction again added no benefit and remained disabled.

M14 reinterpreted frozen covariance primitives as hierarchical Gaussian mixtures. Proper probabilistic fields substantially improved NLL and ECE but not accuracy; for example, sphere multinomial NLL improved from 0.4847 to 0.4772 while accuracy moved by -0.15 points. This reinforced the separation between calibration and classification.

M15 tested auditable fusion of geometric and probabilistic field scores. Global and per-class covariance temperatures and hybrid readouts were fit under a matched five-seed protocol. Hybrid NLL improved over either field-only input within the accuracy tolerance, but a feature-space logistic control remained better. These findings motivated the later use of strong substitutable controls instead of making explicit-field identity the endpoint.

5.9 Phase VIII: E0–E11 systems qualification before research claims

The project next added resumability, content-addressed bundles, compatibility validation, activation, and rollback. This phase did not claim predictive superiority. Its question was whether experiments and model changes could survive interruption and be reproduced exactly.

Failure-injection tests covered:

- class-fitting interruption and partial-class replacement;

- optimizer, momentum, scheduler, scaler, and RNG restoration;
- per-epoch cursor and distributed-sampler restoration;
- incompatible worker-topology rejection;
- corrupted bundle and fingerprint mismatch rejection;
- failed canaries and coordinator loss; and
- deterministic JSON/CSV history export.

These tests established the parent-child transaction model later used for adaptation. They also established a distinction that persists throughout the thesis: a mechanism may qualify operationally without qualifying predictively.

5.10 Phase IX: v5 metric and representation falsification

The v5 methodology introduced versioned stages, deterministic factorial registries, lineage guards, paired statistics, Pareto comparisons, and canonical result artifacts. Ten representation-by-head cells replayed byte-identically, while a deliberate split mismatch failed closed.

Six precision families—spherical, diagonal, full, low-rank, shared, and policy-selected variants—were then tested. The adaptive metric-support policy lost 0.323 balanced-accuracy points and reduced neither parameter bytes nor deterministic fitting work. Both predictive and efficiency gates failed.

This negative result changed the research strategy. Rather than adding more unregistered metric flexibility, the next program locked a strong representation and matched controls, then asked which geometric choices could close a measured predictive gap.

5.11 Phase X: v6 predictive methodology

5.11.1 Locked parent and cheap falsification

M27 froze parent evidence, baseline predictions, teacher lineage, primitive budgets, and three evaluation seeds. The methodology used an S1 cheap falsification stage before the more expensive S2 confirmation stage.

Boundary distillation into component-matched spheres was the first candidate. It reached only 79.80% development accuracy, 6.30 points below the retained explicit

model, and stopped at S1 despite exact replay. This rejected the hypothesis that imitating the RBF boundary with the existing primitive family was sufficient.

5.11.2 Subspaces and directional geometry

Ranked affine-subspace primitives were introduced next. Rank-32 radial scoring reached 89.00%, improving the prior explicit head by 2.90 points and passing the cheap gate. Spherical caps then tested whether directional geometry explained additional error. Cosine caps beat matched Euclidean spheres by 3.97 points over seeds 11, 23, and 37, with a positive seed-level interval.

These results supported two intermediate conclusions: frozen deep features contain useful low-rank and directional structure, and primitive choice matters. They did not establish parity with the discriminative control.

5.11.3 Identifiable factorial and parity stopping rule

M31 crossed objective, primitive, score, and budget factors in an identifiable fractional factorial. The direct rank-32 radial family reached 90.77%, but remained 6.00 points behind the RBF SVM. Because the registered parity operand failed, downstream parity-rescue branches closed.

The v6.1 amendment allowed only bounded, mechanistically motivated tests: tangent caps, constrained nonnegative weights, a component-budget diagnostic, an independent locality frontier, and a separate few-shot feasibility track. Weighted affine readout improved the explicit model to 91.73%, but the final 5.03-point RBF deficit still exceeded the 2-point gate. The predictive program ended with Outcome D rather than extending the search indefinitely.

5.12 Phase XI: lifecycle claims separated from accuracy claims

The A3 edit suite tested 24 explicit transactions. Every transaction rolled back exactly, but unaffected-prediction preservation fell to 84.2% against a 99.9% locality contract. This result showed that exact rollback and local forward behavior are different properties.

The Flowers few-shot track separately tested feasibility. Rank-3 fitting was possible with five examples per class and replayed exactly; rank 32 required 34 examples per class and was declared infeasible rather than estimated from insufficient support. This introduced a further methodological category: *infeasible* is distinct from *failed*.

5.13 Phase XII: v7 stage-wise open-world methodology

With predictive parity closed, the research question changed. The claim was no longer that explicit geometry matches an RBF classifier. It became whether an auditable open-world loop could qualify stage by stage and compose.

M38 preregistered the claim boundary, immutable parents, schedule families, and semantic authority rules. M39 compared six matched rejection heads and retained the low-rank Gaussian, demonstrating that the system claim did not depend on retaining an SDF. M40 compared five grouping policies. M41 tested confirmation- gated updates. M42 tested empirical routing against exhaustive authority. M43 then admitted only stage-passing mechanisms into the integrated factorial.

This sequential retention design reduced researcher degrees of freedom:

1. a mechanism had to pass its local gate before composition;
2. failed routing remained shadow-only;
3. semantic publication always required confirmation;
4. integrated failure blocked untouched confirmation; and
5. stage evidence could not be relabeled as lifecycle evidence.

The final composition failure changed the methodology again. Purity, unknown-recall, and isolated target-gain gates had all appeared successful, but the composed loop integrated no grouped class.

5.14 Phase XIII: v8 endpoint redesign

The v8 program was registered as a new claim rather than a rescue of v6.1 or v7. Its guiding change was to replace per-stage advancement as the primary criterion with post-integration adaptation utility at a fixed review budget.

M45 formalized interface contracts and six fail-closed mismatch tests. M46 tested threshold transfer and audited sufficient statistics. M47 compared six review policies over nine paired cells. M49 independently tested whether parent-scoped fusion repaired locality.

The pivotal design included explicit kill switches. M48 learned selection and M50 untouched lifecycle qualification could run only if M47 passed practical, statistical, safety, and transactional gates. M47 produced a consistent positive effect but failed the practical and safety operands. The downstream branches therefore remained closed.

5.15 Experiment and test ledger

Table 5.1 records the principal hypothesis-driven experiments that led to the final thesis. “Tests” denotes focused regression tests attached to that milestone where recorded; factorial cells and seeds are reported separately because they are experimental observations, not software unit tests.

Table 5.1: Evolution of the principal methodology and tests.

Stage	Question or intervention	Design	Decision
Tier 6	Repair temporal state, calibration, and GPU semantics	WikiText real run and pipeline benchmark	Established 23.91% isolated and 23.35% pipeline baselines; no broad sequence claim.
M0	Freeze protocols, hashes, manifests, and resource counts	Canonical configs and smoke artifacts	Reproducible evidence base; no accuracy claim.
M1	Are primitive semantics and implementations correct?	28 then 34 fast tests	RNG, pruning, CSG gradients, and CPU/GPU parity repaired.
M2	How much error belongs to calibration versus geometry?	5 readouts on one split	Temperature and multinomial readouts sharply improved NLL/ECE.
M3	How does GEODE compare on identical features?	8 baselines $\times$ 5 seeds	GEODE trailed logistic by 0.91 points.
M4	Do subtractive CSG and active repair help causally?	3 ablations $\times$ 5 seeds	Intervals included zero; active repair added no candidates.
M5	Which high-dimensional fitter survives confirmation?	7 fitters; Tier-4 screen and Tier-5 lock	Screen winner reversed on confirmation; current fitter retained.

Stage	Question or intervention	Design	Decision
M6	Can explicit state model temporal context competitively?	Synthetic gates and 50k/10k WikiText	Flexibility shown; linear and n-gram controls won.
M7	Are confidence and OOD scores reliable?	Controlled shift plus CIFAR-100/SVHN	Relative score ranking established; FPR95 limited claim.
M8	Are models robust, stable, and behaviorally editable?	5-seed corruption, stability, edit tests	Broad robustness rejected; tested edits reversible.
M9	How does editability scale?	Class, dimension, and primitive sweeps; 5 repeats	Locality passed; exhaustive and snapshot costs exposed.
M10	Can calibrated scores support open-set inference?	3 locked episodes	No cell passed known and unknown gates jointly.
M11	Can accumulated rejects form safe persistent groups?	40 iterative studies and closure audit	Partial group recovery; semantics and mutation remained human-gated.
M12	Can exact routing become sublinear and faster?	8 scaling, bound, indexing, pruning, and audit iterations	Exact candidates did not satisfy score, work, and tail-latency requirements jointly.
M13	Which primitive family and CSG policy helps?	5-seed CPU screen and 15-cell GPU matrix	Sphere additive retained; subtraction again null.

Stage	Question or intervention	Design	Decision
M14	Do probabilistic field semantics improve calibration?	Matched CPU and OpenCL comparisons	NLL and ECE improved; accuracy did not.
M15	Does geometric/probabilistic fusion help?	5-seed matched readouts; 179-test gate	Hybrid NLL improved; feature logistic remained better.
E0–E2	Can interrupted research resume exactly?	19 focused tests growing to a 187-test gate	Feature, class, optimizer, sampler, RNG, and history state recovered exactly.
E3–E4	Can bundles and a real CIFAR lifecycle qualify?	192 then 196 tests	Bundle lifecycle passed; CIFAR non-inferiority passed with weak near-OOD FPR95.
E7	Can distributed DomainNet execution qualify?	3 logical nodes; 192-image episode	Local recovery passed; physical multi-host gate remained blocked.
E10–E11	Can serving recover and public artifacts reproduce?	Fault rehearsal and 30 frozen inputs	1-second RTO/zero-request RPO passed; five outputs reproduced byte-identically.
M16	Can the protocol replay and reject split drift?	10 cells; 15 tests	Byte-identical replay; deliberate split mismatch rejected.
M18	Do adaptive precision families improve accuracy or cost?	6 families; 13 tests	Lost 0.323 points; no resource reduction; stopped.
M27	Can v6 begin from immutable matched parents?	3 heads; 254-test full gate	Parents, predictions, teacher, and budgets locked.

Stage	Question or intervention	Design	Decision
M28	Can RBF boundaries be distilled into matched spheres?	S1; 4 tests	79.80%; 6.30 points below parent; stopped.
M29	Do low-rank affine subspaces help?	Rank sweep; 6 tests	Rank-32 radial gained 2.90 points; advanced.
M30	Does directional geometry matter?	3 seeds; 6 tests	Cosine exceeded Euclidean by 3.97 points; mechanism retained.
M31	Which factor family approaches parity?	9 cells $\times$ 3 seeds	90.77%, 6.00 points behind RBF; parity branches closed.
A1-T	Do tangent-space caps rescue the model?	S1; 10 tests	Worse than affine control; stopped.
A1-W	Does constrained component weighting help?	S1; 10 tests	Added 2.60 points and reduced NLL; advanced.
A1-B	Is more component capacity monotonic?	8 tests	Peaked then declined; classified as unstable.
A2	Does weighting close parity on all seeds?	3 seeds; 7 tests	91.73%, 5.03 points behind RBF; Outcome D.
A3	Are edits local and reversible?	24 transactions; 7 tests	Rollback exact; locality only 84.2%.
A4	Is five-shot fitting feasible?	2 rank tiers; 4 tests	Rank 3 feasible; rank 32 blocked by sample count.
v6.1 final	Can conclusions replay without source data?	7 indexes, 65 artifacts; 5 tests	11 conclusions reproduced byte-identically.

Stage	Question or intervention	Design	Decision
M38	Are v7 claims, parents, and schedules frozen?	3 schedule families; 10 tests	Protocol and prior-art boundary locked.
M39	Which rejection head passes matched gates?	6 heads $\times$ 3 seeds	Only rank-32 Gaussian retained.
M40	Which grouping policies create stable review objects?	5 policies; 15 cells	HDBSCAN primary; FINCH control.
M41	Which confirmed updates are safe?	24 operation cells	New-class rank-16 insertion retained.
M42	Can empirical routing become authoritative?	4 profiles; 12 cells	No; routing remained shadow-only.
M43	Do retained stages compose?	36 factorial cells	Grouped arms integrated 0/3 classes; Outcome C.
v7 final	Can Outcome C replay artifact-only?	6 indexes, 16 artifacts	8 conclusions reproduced twice.
M45	Can interfaces fail closed?	7 schemas; 6 mismatch cases; 11 tests	All registered mismatches rejected.
M46	Do thresholds transfer and cores represent support?	4 rules; 3 seeds; 5 tests	Anchor quantile retained; core support defect identified.
M47	Does utility selection beat core at equal budget?	6 policies; 54 arm cells; 3 tests	Positive 9/9, but practical and safety gates failed.
M49	Can parent-scoped residuals repair locality and utility?	2 attempts; 2 tests	100% locality; no residual retained.

Stage	Question or intervention	Design	Decision
v8 final	Can Outcome D replay artifact-only?	4 indexes, 11 artifacts; 1 test	6 conclusions reproduced twice without data access.

5.16 Test taxonomy

The test count alone does not characterize the evidence. The final 374-test repository gate combines several distinct test classes:

Semantic unit tests verify equations, normalization, gradients, shapes, and deterministic RNG behavior.

Property and invariant tests verify conservation, orthonormality, monotonic fail-closed rules, graph consistency, and immutable class order.

Parity tests compare scalar and vectorized implementations, CPU and GPU paths, and pre- and post-serialization predictions.

Failure-injection tests interrupt fitting, corrupt bundles, alter fingerprints, remove confirmation, stale thresholds, exceed budgets, or lose workers and coordinators.

Leakage tests reject split mismatch, partition overlap, use of validation examples as adaptation support, and final-label access.

Lifecycle tests exercise publication, child lineage, pointer changes, rollback, stale-profile fallback, and audit history.

Statistical tests implement paired comparisons, bootstrap intervals, seed consistency, and practical effect gates.

Artifact-replay tests recompute frozen conclusions from indexed JSON without loading training data or opening sealed labels.

Software tests establish implementation and evidence integrity; they do not replace experimental replication. Conversely, repeated experimental cells do not prove that transaction and leakage invariants are implemented correctly. The methodology deliberately requires both.

5.17 How negative results shaped the final design

The main methodological changes can be traced directly to failures:

Observed failure	Resulting methodological change
CSG subtraction was null or negative	Keep subtraction opt-in and require causal matched ablations.
Adaptive metric support reduced neither error nor cost	Stop adding metric flexibility without a registered utility hypothesis.
Explicit heads remained behind RBF	Treat acceptance heads as substitutable; separate editability from predictive superiority.
Rollback passed while locality failed	Register locality independently and freeze affected regions from the parent.
Strong stages failed to compose	Replace stage metrics with adaptation utility as the primary endpoint.
Pure cores failed adaptation	Audit producer-consumer sufficient statistics and compare representative review policies.
Utility gain harmed unknown recall	Make remaining-unknown recall a binding lifecycle constraint.
Positive interval missed practical threshold	Preserve Outcome D and block post-hoc learned selection and untouched confirmation.

5.18 Methodological maturity

The methodology matured from exploratory implementation to preregistered falsification. Early experiments discovered semantics and established baselines. Later programs froze parents, controls, budgets, seeds, effect-size thresholds, safety constraints, and branch dependencies before evaluation. This evolution reduces flexibility but strengthens the interpretation of a negative result.

The final thesis is therefore not a report of one successful algorithm. It is a documented sequence of hypotheses in which each retained conclusion is tied to a testable gate, and each stopped branch remains part of the scientific record.

Chapter 6

Stage-Wise Open-World Study

6.1 Data and representation

The realized experiment uses immutable 384-dimensional CLS-token features from DINOv2-small [29] on CIFAR-10. Each frozen seed contains 10,000 training and 1,000 development examples. Seeds are 11, 23, and 37. In the v7 leave-two-class-out protocol, classes 0–7 are known, class 8 is confirmable, and class 9 remains unknown.

Broader CIFAR-100 and DomainNet tracks described in earlier planning were not executed in this chain and are not presented as evidence. Final confirmation labels remained sealed after the integrated gate failed.

6.2 Acceptance-head comparison

Six heads share the same representation and fitting data:

1. multinomial maximum posterior;
2. 1-nearest-support distance;
3. rank-32 class-conditional Gaussian;
4. EVM-style Weibull margin approximation;
5. weighted affine support; and
6. sigmoid-calibrated RBF SVM evidence.

A stratified train-only calibration split selects the novelty threshold at approximately 92% known coverage. Retention requires at least 90% known coverage and 50% unknown recall on every seed, bounded known-accuracy loss, competitive review precision, corruption checks, and exact replay.

6.3 Persistent discovery

Rejected examples enter a bounded FIFO buffer of 2,000 records. Five deterministic windows compare no clustering, streaming centroid micro-clusters, HDBSCAN, FINCH, and silhouette-selected k -means. Groups need minimum size 10 and at least 70% purity for scoring. A review ID persists when consecutive-window member overlap is at least 50%.

Discovery advancement requires category recovery, review precision, stable identities, replay, bounded resources, and no semantic publication.

6.4 Confirmation and adaptation

A deterministic oracle simulates delayed human confirmation while preserving the authority boundary. Outcomes are existing class, new class, corruption/irrelevant, or unresolved. Operations include no update, native Gaussian update, rank-16 affine insertion, full class-local refit, and a full-retraining control.

New-class adaptation uses 50 confirmed supports. Passing requires at least 5 points of target gain, at most 1 point of known regression, at most 2 points of remaining-unknown loss, stable NLL, graph validity, confirmation lineage, replay, and rollback.

6.5 Routing

Three independently owned bundles have class orders $\{0, 1, 2, 3\}$, $\{4, 5, 6, 7\}$, and $\{8\}$. Profiles include centroids and radii, rank-16 Gaussians, compact prototypes, and undercomplete autoencoders. Structural fingerprints determine compatibility. Empirical profiles shortlist two bundles.

Authority requires at least 99% correct-bundle top-1, 99.9% exhaustive-winner inclusion and final agreement, 95% no-confident-route recall, 25% exact-model evaluation reduction, lower tail latency, exact update replay, and complete stale-profile fallback.

6.6 Integrated factorial

Only retained mechanisms enter the integrated study: rank-32 Gaussian rejection, HDBSCAN or FINCH discovery, delayed confirmation, rank-16 insertion, and exhaustive routing. A no-grouping review-everything arm is retained as a burden control. The grouped review budget is 50.

The integrated gate requires at least 75% of confirmable classes integrated, 25% review reduction, known and unknown preservation, no unconfirmed publication, and all transaction contracts.

Chapter 7

Adaptation-Utility Study

7.1 Episode replay

The v8 study begins with classes 0–5 and introduces classes 6, 7, and 8 in order for each of the three seeds, producing nine seed-by-arrival cells. Class 9 remains unknown. Each class is deterministically partitioned into geometry, anchor, review-candidate, and proxy-validation subsets.

The proxy-validation subset scores a frozen candidate review set but never becomes adaptation support. Each episode records partition hashes, class order, selected IDs, confirmation IDs, thresholds, parent and child hashes, utility, safety metrics, and rollback evidence.

7.2 Threshold-transfer experiment

Adding a class changes score competition. Four rules are compared:

1. frozen pre-integration threshold;
2. class-count heuristic;
3. global anchor-quantile recalibration; and
4. per-class anchor thresholds under a global fail-closed ceiling.

The experiment consumes no review labels. Stale anchor lineage never lowers confidence and restores the parent threshold. The selected rule must preserve known accuracy and remaining-unknown recall over all three seeds.

7.3 Sufficient-statistics audit

The audit measures whether discovery outputs represent the candidate region. Frozen features include:

- distance to cluster center;
- distance to the parent decision boundary;
- low-rank subspace coverage;
- covariance trace ratio;
- omitted-region nearest-neighbor coverage; and
- class-mode coverage.

The audit also ranks measured interface mismatches by failure frequency, estimated utility, and implementation cost.

7.4 Equal-budget review policies

All eligible arms receive 50 labels:

Core-selected: nearest examples to the candidate center.

Boundary-inclusive: deterministic mixture of core, modes, and parent-boundary quantiles.

Coverage-selected: farthest-first k -center support.

Utility-selected: deterministic subsets ranked by a rank-3 proxy adapter on the disjoint validation partition.

Random-stratified: seeded sampling over two candidate modes.

Review-everything: all 300 candidates; diagnostic and ineligible.

The production adapter remains rank 16. All candidates in this controlled study belong to the arriving class, fixing purity at 100% and isolating representativeness.

7.5 Statistical analysis

The primary comparison is paired by seed and arrival class. Mean utility difference is accompanied by a deterministic 10,000-resample paired percentile bootstrap interval. Cell signs are reported because a mean can conceal schedule-specific failures. Advancement requires practical effect, interval, cell consistency, and safety jointly.

7.6 Local residual branch

Two residual families are preregistered:

1. local target-class temperature; and
2. bounded class-local affine correction.

The parent freezes the affected region before fitting using selected support, a bounded nearest-neighbor radius, and component responsibility. Fusion is gated by this parent-only mask. Passing requires 5 points of utility gain, 99.9% unaffected-prediction preservation on every seed, known and unknown safety, local update work, replay, and rollback. At most two attempts are allowed.

Chapter 8

Results

8.1 Predictive competitiveness of explicit heads

The v6.1 weighted rank-32 affine head reaches 91.73% balanced accuracy, improving its retained explicit predecessor by 0.97 points. The matched RBF SVM reaches 96.77%, leaving a 5.03-point gap and failing the registered 2-point parity operand. A direct radial family peaks and then deteriorates as component budget increases, indicating instability rather than monotonic capacity gain.

Table 8.1: Selected predictive results under frozen DINOv2 features.

Head	Balanced accuracy	Difference from RBF
Weighted affine rank 32	91.73%	−5.03 points
Direct radial rank 32	90.77%	−6.00 points
RBF SVM control	96.77%	—

All lifecycle and replay gates pass for the selected explicit head. The result answers RQ1 negatively for predictive parity but positively for deterministic, editable artifact behavior. Editability is not evidence of accuracy leadership.

8.2 Acceptance-head results

The low-rank Gaussian is the only v7 acceptance head to pass every autonomy and review gate.

This result reinforces the distinction between geometric coverage and calibrated discrimination. Proper density is retained over raw affine support.

Table 8.2: Three-seed mean acceptance-head results.

Head	Coverage	Unknown recall	AUROC	Review precision	Retained
Max posterior	91.38%	60.17%	0.9053	69.33%	No
kNN support	91.83%	67.00%	0.9045	86.67%	No
Low-rank Gaussian	92.08%	86.17%	0.9561	98.67%	Yes
EVM-style margin	100.00%	0.00%	0.5102	22.00%	No
Weighted affine	90.75%	53.00%	0.8763	64.67%	No
RBF evidence	93.08%	12.83%	0.4010	31.33%	No

Table 8.3: Persistent discovery results.

Policy	Group recall	Review precision	Full cells	Pass
No clustering	0.00%	0.00%	0/9	No
Micro-clusters	0.00%	98.67%	0/9	No
HDBSCAN	100.00%	100.00%	9/9	Yes
FINCH	83.33%	86.39%	8/9	Yes
<i>k</i> -means	100.00%	98.67%	9/9	Yes

8.3 Discovery results

HDBSCAN and FINCH produce stable review objects and pass stage gates. The micro-cluster result shows that sample-level review precision does not imply separation of distinct unknown classes.

8.4 Isolated adaptation and routing

Confirmed rank-16 insertion raises held-out class-8 success by 42%, 52%, and 36% across seeds, a 43.33-point mean gain. Known accuracy and remaining-unknown recall remain within bounds, and all child bundles replay and roll back exactly.

No routing profile qualifies for authority. Best top-1 is 93.53%, best winner inclusion is 99.57%, and best unknown fallback recall is 71.00%. Profiles reduce exact model evaluations by approximately 50%, but compute savings are not a safety result. Routing remains shadow-only.

8.5 Stage composition failure

Qualified HDBSCAN and FINCH loops integrate no confirmable classes. Reviewing all rejections integrates all three, but eliminates the human-budget advantage. Remaining-unknown recall is 6.83 points below the M39 stage baseline. All confirmation, graph, replay, rollback, and audit contracts pass. The failure is therefore one of utility and composition, not transaction correctness.

Table 8.4: Integrated v7 delayed-confirmation results.

Discovery	Integrated	Reviews	Review reduction	Unknown recall
HDBSCAN	0/3	50.00	78.78%	79.33%
FINCH	0/3	15.33	93.49%	79.33%
Review everything	3/3	235.67	0.00%	79.33%

8.6 Threshold transfer

Frozen, class-count, and global anchor-quantile rules all preserve mean known accuracy and remaining-unknown recall in the isolated M46 episodes. The global anchor rule is retained because it directly recalibrates after class-order change. Per-class thresholds increase unknown rejection but cause 1.875 points of mean known regression, exceeding the one-point ceiling.

Table 8.5: Threshold transfer over three seeds.

Rule	Known drop	Unknown drop	Pass
Frozen parent	0.000	0.000	Yes
Class-count heuristic	0.000	0.000	Yes
Global anchor quantile	0.000	0.000	Yes
Per-class anchor	1.875	−8.667	No

8.7 Representativeness audit

Table 8.6: Core and boundary-inclusive support diagnostics.

Seed	Core NN cov.	Boundary NN cov.	Core trace ratio	Boundary trace ratio
11	34.04%	99.68%	0.663	1.078
23	35.47%	99.78%	0.658	1.096
37	39.14%	99.37%	0.679	1.068

Core and boundary sets both cover two coarse modes, so mode count alone misses the defect. Nearest-neighbor support and covariance reveal that core sets omit most of the rejected region. Boundary-inclusive sets nearly recover full support, although the cheap M46 proxy utility improves by only 0.259 points.

Table 8.7: Mean v8 adaptation utility over nine cells.

Review policy	Mean U_e	Labels	Remaining-unknown recall
Core-selected	6.36	50	59.72%
Boundary-inclusive	9.71	50	75.98%
Coverage-selected	10.62	50	82.50%
Utility-selected	9.96	50	78.15%
Random-stratified	9.70	50	75.41%
Review-everything	11.26	300	86.13%

8.8 Equal-budget utility

Utility selection beats core selection in all nine paired cells. The mean difference is 3.593 points with paired 95% bootstrap interval [2.794, 4.382]. The sign and interval operands pass, but the 5-point practical operand fails. Six cells also exceed the 2-point remaining-unknown-recall band.

Table 8.8: Pivotal preregistered gate.

Operand	Observation	Pass
Mean gain over core ≥ 5.0	3.593	No
Paired 95% lower bound > 0	2.794	Yes
Positive cells $\geq 7/9$	9/9	Yes
Budget, safety, transaction conjunction	6 recall violations	No

Coverage selection is the strongest eligible deterministic arm, slightly outperforming utility selection. The rank-3 proxy therefore does not perfectly rank rank-16 production outcomes. Review-everything achieves the highest raw utility but is ineligible at six times the budget.

8.9 Local residuals

Both residual attempts preserve 100% of unaffected predictions, improving on the earlier A3 minimum of 84.2%. The local temperature adds 0.000 points of mean utility. The bounded affine correction adds 0.204 points. Maximum known regression is 0.125 points, but maximum remaining-unknown-recall drop is 4.333 points. Neither residual is retained.

This result separates locality from utility. Explicit parent-scoped fusion repairs the scope bug, but a correctly localized update is not necessarily a useful or safe update.

8.10 Final disposition

M47 produces registered Outcome D. Learned selection, same-harness head continuation, and untouched M50 lifecycle qualification are blocked. The positive bootstrap interval is reported as a diagnostic finding, not promoted to primary success.

Chapter 9

Discussion

9.1 Answer to RQ1

Compact explicit geometry did not reach predictive parity with the matched RBF control. The weighted affine head remained 5.03 points behind. Proper class-conditional Gaussian density also outperformed weighted affine support for rejection. These outcomes do not remove the governance value of explicit artifacts, but they reject the premise that editability itself implies competitive discrimination.

9.2 Answer to RQ2

Independently qualified stages did not compose. HDBSCAN achieved perfect stage-wise group metrics, and confirmed insertion delivered a large isolated gain, yet their grouped loop integrated no class. The review-everything control showed that the adapter could succeed when given broader support. The failure therefore lies at the clusterer-to-review-to-adapter interface.

9.3 Answer to RQ3

Selecting for downstream utility improves over core selection consistently: 9/9 cells and a positive bootstrap interval. The improvement is nevertheless only 3.593 points and violates remaining-unknown safety in six cells. The answer is therefore qualified: adaptation utility is a more discriminating endpoint and reveals a real representativeness signal, but the implemented utility selector does not qualify the lifecycle.

9.4 Answer to RQ4

Transactional controls successfully distinguish predictive failure from systems failure. No unconfirmed class is published, every rollback restores the exact parent, stale lineages fail closed, and residual scope can be enforced to preserve all unaffected predictions. These controls do not create accuracy, but they make negative results attributable and reproducible.

9.5 Pure cores as insufficient statistics

Density-based discovery naturally prefers high-cohesion regions. Such cores are attractive review objects because they reduce ambiguity and label noise. The adapter, however, estimates a class region. It needs variance directions, boundary examples, and low-density modes. The measured 34–39% support coverage provides a direct explanation for the v7 failure.

This distinction generalizes beyond the specific adapter. A prototype learner, density estimator, margin model, or discriminative head can require different support statistics. Review selection should therefore be conditioned on the downstream update mechanism.

9.6 Adaptation and unknown preservation conflict

Broader support makes the new class easier to recognize, but it also expands acceptance into regions occupied by future unknowns. This creates a structural tradeoff. Utility on known plus discovered classes and recall on remaining unknowns cannot be optimized independently.

A future objective should be explicitly constrained:

$$\max_{\pi} \mathbb{E}[U_e(\pi)] \quad \text{subject to} \quad \Delta \text{BA}_{\text{known}} \geq -\epsilon, \quad \Delta R_{\text{unknown}} \geq -\delta, \quad \text{cost}(\pi) \leq B. \quad (9.1)$$

Here π is the review policy. Unknown preservation is not a post-hoc diagnostic; it is part of the optimization problem.

9.7 Why coverage beats the utility proxy

The utility selector evaluates deterministic subsets with a rank-3 proxy, then deploys a rank-16 production adapter. This mismatch introduces proxy regret. Farthest-first

coverage has no such model mismatch and reaches 10.62 points, compared with 9.96 for utility selection.

Possible improvements include a better-fidelity cheap adapter, uncertainty in the subset ranking, direct constrained selection, or a learned policy trained across episodes. The registered kill switch correctly blocks these continuations after the pivotal endpoint fails; they remain future hypotheses.

9.8 Methodological implications

The results support four methodological recommendations:

1. Register a system-level endpoint before optimizing stages.
2. Audit producer outputs against consumer sufficient statistics.
3. Treat unknown recall as a lifecycle state variable after every update.
4. Separate transaction correctness, locality, and predictive utility.

Chapter 10

Validity, Limitations, and Ethics

10.1 Internal validity

The use of frozen partitions, seeds, parent hashes, and deterministic replay reduces accidental variation. Paired cells isolate policy differences. The validation partition is disjoint from adaptation support. Fail-closed tests guard common lineage and leakage errors.

Internal validity remains limited by the deterministic oracle and controlled candidate purity. Real review candidates may mix classes, and real reviewers may disagree. The utility proxy and production adapter differ in rank.

10.2 Construct validity

Balanced accuracy gives equal weight to classes and is appropriate for the episode inventory, but it does not capture calibration, severity-weighted errors, or real deployment value. The fixed 50-label budget approximates human cost but does not model variable review difficulty.

Nearest-neighbor support coverage and covariance ratios diagnose representativeness but are not complete sufficient statistics. Their role is explanatory rather than universal.

10.3 External validity

The realized study uses CIFAR-10 and frozen DINOv2-small features. Results may change with more classes, long-tailed data, domain shift, natural streams, larger encoders, or representation adaptation. The held-out class order is specific. The study does not establish performance for medical, legal, security, or other high-stakes applications.

10.4 Statistical conclusion validity

There are nine paired development cells. The bootstrap interval describes this bounded schedule, not a broad population of domains. The untouched M50 schedule remains sealed, so there is no independent final confirmation. The preregistered practical threshold prevents a positive interval from being overinterpreted.

10.5 Ethics and governance

Open-world systems can assign meaning to coherent but irrelevant, private, or harmful groups. Automated class creation may amplify surveillance, discrimination, or feedback loops. This work prohibits autonomous semantic publication and records review authority, provenance, and rollback.

Deployment would additionally require privacy-aware retention, deletion and appeal procedures, reviewer disagreement models, subgroup audits, access control, security review, and monitoring for distributional feedback. The experiments use public benchmark representations and simulated review; they do not involve human subjects or sensitive personal data.

10.6 Claim boundary

The evidence supports the claims that:

- adaptation utility is more discriminating than stage purity;
- boundary- and coverage-aware review improves utility over density cores in the frozen harness;
- global anchor recalibration transfers safely in the isolated M46 episodes; and
- parent-only fusion repairs the observed locality leakage.

It does not support claims of closed-set parity, SDF necessity, autonomous class creation, learned-selector superiority, authoritative routing, head-substitution superiority, untouched confirmation, or qualified end-to-end operation.

Chapter 11

Reproducibility

11.1 Artifact structure

The repository stores implementation code, tests, plans, claim ledgers, machine-readable evidence, and immutable artifact indexes. Key locations are:

- `analysis/V8_FINAL_CLAIM_LEDGER.md`;
- `analysis/RESEARCH_IMPLEMENTATION_PLAN_v8.md`;
- `logs/results/v8/m46_diagnostics/`;
- `logs/results/v8/m47_review_utility/`;
- `logs/results/v8/m49_local_residuals/`; and
- `logs/results/v8/final_outcome_replay/`.

11.2 Final replay

From the repository root on Windows:

```
.\.venv\Scripts\python.exe -m experiments.tier1.eval_v8_final_replay
```

The command verifies four immutable milestone indexes, 11 indexed artifacts, and six conclusion operands. It runs twice and requires byte-identical output, loads no training data, and opens no final labels. The complete repository gate contains 374 tests at finalization.

11.3 Building this report

From the `analysis` directory:

```
tectonic MS_THESIS_REPORT.tex
```

Tectonic resolves standard LaTeX packages automatically. The source uses an in-line bibliography and no external figures, reducing archival dependencies. Institution-specific title, approval, declaration, margins, and binding rules must be adapted before university submission.

Chapter 12

Conclusions and Future Work

12.1 Conclusions

This thesis investigated open-world recognition as a composed, transactional lifecycle. The program first tested whether explicit geometric artifacts could serve as competitive class heads. They replayed and rolled back reliably but remained behind discriminative controls. It then qualified rejection, discovery, and confirmed adaptation stages independently. Those stages failed when composed.

The successor study replaced stage metrics with adaptation utility at a fixed review budget. It found that density cores omit most of the candidate region and that representative selection improves downstream utility. The positive effect is consistent but insufficient: it misses the registered practical threshold and damages remaining-unknown recall.

The principal conclusion is methodological. Open-world learning is not a pipeline whose stages can be optimized independently. Every update changes the known inventory, acceptance threshold, future unknown space, and routing state. Qualification must therefore occur at the lifecycle level under joint utility, safety, cost, lineage, locality, and rollback constraints.

12.2 Future work

The next research program should begin with new registration rather than reopen Outcome D. Promising directions are:

1. constrained review selection that predicts both adaptation gain and remaining-unknown loss;
2. higher-fidelity but bounded proxy adapters;
3. leave-one-episode-out selection learners with explicit uncertainty;

4. evaluation over larger category inventories and natural class-arrival streams;
5. real human-review studies with disagreement and variable cost;
6. representation adaptation under strict parent and rollback contracts;
7. mechanism-matched comparisons among Gaussian, EVM, kNN, and explicit geometry under the same lifecycle endpoint; and
8. router training that treats unknown fallback as a primary safety target.

Any continuation should retain the central discipline of this work: practical effect thresholds and safety constraints must be registered before tuning, and failed branches must remain visible.

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Appendix A

Registered Gates

Stage	Principal advancement conditions
Predictive head	Within 2 points of RBF control, lifecycle checks, exact replay and rollback.
Rejection	Every seed: at least 90% known coverage, 50% unknown recall, bounded known loss, competitive review precision, exact replay.
Discovery	Category recovery, at least 70% precision, stable review IDs, resource limits, no semantic publication.
Adaptation	At least 5 points target gain, at most 1 point known drop, at most 2 points unknown-recall drop, confirmation, replay, rollback.
Routing	At least 99% top-1, 99.9% winner inclusion and agreement, 95% unknown fallback, and 25% compute reduction.
Integrated v7	At least 75% classes integrated, 25% review reduction, known/unknown preservation, all safety contracts.
Utility v8	At least 5 points over core, positive paired interval, at least 7/9 positive cells, budget and safety conjunction.
Residual	At least 5 points utility, 99.9% unaffected preservation per seed, known/unknown safety, local work, replay, rollback.

Appendix B

Final Branch Disposition

Branch	Status	Reason
Weighted affine parity	Failed	91.73%, 5.03 points behind RBF.
Gaussian rejection	Passed	Only head satisfying all v7 acceptance gates.
HDBSCAN discovery	Passed stage	Perfect stage-wise group recall and precision.
FINCH discovery	Passed stage	Mechanistically distinct retained control.
New-class insertion	Passed stage	43.33-point isolated target gain with rollback.
Authoritative routing	Closed	Accuracy and unknown-fallback gates missed.
Integrated grouped loop	Failed	0/3 classes integrated.
M46 threshold transfer	Passed	Global anchor quantile retained.
M47 utility selection	Outcome D	3.593 points below 5-point gate; six safety violations.
M48 learned selector	Blocked	Binding M47 kill switch.
M49 residuals	Closed	Locality passed; utility and unknown safety failed.

Branch	Status	Reason
M50/E12 confirmation	Blocked	Untouched schedule remains sealed.

Appendix C

Artifact-Only Verification

The final v8 verifier reports:

- four verified milestone indexes;
- 11 verified indexed artifacts;
- six reproduced conclusion operands;
- byte-identical output across two runs;
- no training-data loading; and
- no final-label access.

The final claim ledger is `analysis/V8_FINAL_CLAIM_LEDGER.md`. The paper-length presentation is `analysis/FINAL_RESEARCH_PAPER.tex`. This thesis report provides the expanded academic narrative; it does not introduce new experimental claims.